Do Synthetic Personas Predict Real Conversion? A Blind, Leakage-Controlled Backtest Across Three Decision Framings

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Abstract

Synthetic personas are increasingly proposed as a cheap substitute for real demand validation, yet their ability to predict real-world conversion is seldom tested against ground truth. We build a blind, leakage-controlled backtest—five documented historical validation episodes, anonymized so an LLM cannot recall their outcomes—and have persona agents from a population-grounded dataset make purchase decisions, scoring against the real result. Across 200 personas per case we find naive one-shot persona simulation is anti-predictive: real successes receive lower simulated conversion than real failures (success−failure separation $\Delta = -0.46$). Crucially, this failure is framing-invariant: reformulating the decision from stated willingness to a longitudinal revealed-preference judgment collapses every response to a floor ($\Delta = 0$), and a graded-probability formulation restores resolution but not discrimination ($\Delta = -0.05$); the case ranking is identical across all three. The failure is also aggregation-invariant: real successes are dominated by failures not only in the mean but at every quantile including the top decile and the maximum, so no hidden enthusiast minority explains the real winners. Applying the method to two real pre-launch products reproduces the bias (predicting $\sim$14% paid conversion for a product whose team targets 5%). Yet the bias is partly correctable: leave-one-out calibration on real outcomes raises directional accuracy from 0.40 (raw) to 0.64 over 14 cases, though a perfect result on the initial five did not survive expansion—the sim–real map is non-monotonic, since some products (over-hyped hardware) are correctly rejected by the personas. We conclude synthetic-persona sims must not be used raw and only partly recover under simple calibration, and we release the blind backtest as a reusable validity gate.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

AI coding lets founders ship faster than ever, yet few products sustain value; demand validation (interviews, fake-door tests, A/B) is the recommended hedge but is costly for solo builders. Synthetic personas promise to automate it, and recent work offers population-grounded persona datasets (NVIDIA, 2025) and multi-agent simulation engines (MiroFish contributors, 2025). The unanswered question is epistemic: can a persona simulation predict conversion that holds in the real world? We contribute (i) a blind, leakage-controlled backtest protocol isolating predictive signal from LLM memorization via case anonymization; (ii) a three-framing robustness study (stated, revealed, graded) showing the failure is not an artifact of one prompt; and (iii) a cautionary real-product application to two live pre-launch products; and (iv) a calibration analysis showing the bias is partly correctable (leave-one-out accuracy rises from 0.40 to 0.64 over 14 cases) but that the sim–real map is non-monotonic, so simple calibration is insufficient. The raw finding is negative, obtained without tuning toward any desired outcome.

2. Related Work

Silicon sampling (Argyle et al., 2023) conditions an LLM on a respondent's backstory to reproduce survey distributions, and now underlies most "AI panel" products. Its validity is uneven: reviews of silicon-to-human comparisons find only a minority match while most diverge, and willingness-to-pay studies report directional alignment in familiar categories but weak performance in novel ones (Anonymous, 2026a;b).

The documented weak spots—novel-category behavior, minority-opinion tails, and actual purchase in unfamiliar contexts—are precisely where demand validation operates. Population-grounded datasets (NVIDIA, 2025) and multi-agent engines (MiroFish contributors, 2025) scale personas, but prior work largely evaluates plausibility or survey-distribution match. We instead test external predictive validity against real conversion outcomes under a blind, leakage-controlled protocol, and quantify how far light calibration closes the gap—corroborating and sharpening the WTP-in-novel-categories weakness.

3. Method

Personas. We sample personas from a ~1M-persona population-grounded Korean dataset (NVIDIA, 2025) (occupation, education, region, consumption narrative), rendered to a compact profile. A local 7B instruction-tuned LLM (Qwen Team, 2024) is prompted to become the persona.

Three decision framings. Stated: one-shot "would you pay?" (binary). Revealed: a longitudinal judgment—click, try, and still pay after 2–8 weeks once novelty fades and incumbents are considered (binary). Graded: a continuous probability of becoming a retained paying customer. A case's simulated conversion is the mean over its personas.

Persona conditions. General: random personas. Target: personas keyword-filtered to each product's target segment, fixed a priori from the product description (never the outcome).

Baseline and leakage control. A no-persona baseline asks the LLM for the conversion rate directly. Each case is anonymized (brand, company, era removed) so the model cannot recall the historical result; ground truth is revealed only for scoring. We report directional accuracy and separation $\Delta = \overline{\text{sim}}_{\text{success}} - \overline{\text{sim}}_{\text{failure}}$ ($\Delta > 0$ predictive, $\Delta < 0$ anti-predictive).

4. Results

Table 1 gives per-case simulated conversion under the graded framing; real successes (A,B,E) sit below real failures (C,D). Table 2 shows this is framing-invariant: no framing or persona condition yields $\Delta > 0$.

Directional accuracy never exceeds the no-persona baseline (0.40). The stated framing over-predicts appealing-but-failed cases; the revealed framing is so strict that every persona declines (a degenerate floor); the graded framing restores resolution but leaves failures ranked above successes. The invariance of the ranking across framings, persona conditions, and against the baseline indicates a structural, not prompt-specific, failure.

Table 1. Graded per-case conversion (200 personas, general). Successes A,B,E score at or below failures C,D. Ranking C>D>B>A>E is identical under all three framings.

Case	Method	GT	Graded
A	fake-door video/waitlist	1	0.153
B	concierge fake store	1	0.202
C	reformulate flagship	0	0.230
D	Wizard-of-Oz	0	0.180
E	landing-page pricing	1	0.126

Table 2. Separation Δ across three framings and two persona conditions. Every value is ≤ 0; no configuration is predictive. Revealed collapses to a floor (all responses 0).

Framing	General Δ	Target Δ
No-persona baseline	−0.233	
Stated (binary)	−0.462	−0.494
Revealed (binary)	0.000	0.000
Graded (continuous)	−0.045	−0.054

Aggregation robustness. One might suspect the mean hides an intense niche minority for the real successes. It does not: recomputing per-case distributions, the failures dominate the successes on the high-intensity fraction, the top-decile mean ($\Delta = -0.21$), the standard deviation, and the maximum ($\Delta = -0.32$). No moment of the persona distribution is predictive, closing the "wrong statistic" objection.

Real-product application. We apply the graded method to two live pre-launch products (anonymized): P1, a real-time civic-debate app with a paid subscription tier, and SoloSquad (SoloSquad, 2026), an open-source multi-agent tool for solo founders. Predicted paid conversion is 0.14 (P1) and 0.10 (SoloSquad), essentially unchanged under target conditioning. P1's own team targets a 5% premium conversion; the simulation over-predicts by ~3×, reproducing the abstract-appeal bias on real inputs.

Calibration partly recovers prediction. The bias is systematic: real successes sit consistently below real failures ($\Delta = -0.18$ over 14 cases). We exploit this with leave-one-out calibration—a one-feature decision stump fit on held-out cases. On the initial five cases LOO reaches 5/5, but expanding to 14 (adding well-documented successes and failures) drops LOO to 0.64—still above raw (0.40) and chance (0.50), yet far from perfect. The residual errors are informative: over-hyped expensive hardware is correctly rejected by the personas (low sim, real failure), breaking the anti-correlation and making the sim–real map non-monotonic. Simple sign-flip calibration therefore helps but does not solve the problem; adding an a-priori price/hype regime feature does not improve LOO (also 0.64), so the ceiling appears set by data scale, not feature choice—reliable validity would require a substantially larger labeled corpus. Pre-launch products without real labels (P1, SoloSquad) still require a real anchor (e.g., a fake-door test) to calibrate.

5. Discussion: Why It Fails

(1) Abstract-appeal bias. One-shot willingness elicits stated appeal, not revealed conversion; broadly-appealing-but-failed products (a sweeter drink, effortless dictation) score high. (2) Missing temporal/relational dynamics. The real failures hinge on brand attachment and sustained-use fatigue—invisible to a single decision; forcing a longitudinal binary instead floors everything. (3) Counterintuitive niche demand. Real successes had low general appeal but strong niche demand, which target filtering and graded probabilities still underrate. Because the same ranking survives every reformulation, the limitation is intrinsic to persona willingness, not to a particular prompt. The bias is systematic enough that calibration against real outcomes (Section 4) partly recovers prediction, but a non-monotonic sim–real map—some duds are correctly rejected— limits any simple correction: the practical recommendation is to treat persona simulations as a feature requiring regime-aware calibration, never a raw estimate.

6. Limitations

Five backtest cases yield low statistical power; we frame results as a mechanism-focused, framing-robust case study, not a definitive effect size. We use one 7B model, one (Korean) persona market, and keyword target definitions; larger models, multi-turn deliberation, and revealed-behavior traces are untested and are exactly what our diagnosis motivates. The real-product predictions have no ground truth and are reported only to illustrate the bias.

Impact Statement

This work cautions against substituting synthetic-persona simulations for real user evidence in product decisions; over-trust could bias which products get built and entrench dataset demographics. By reporting a rigorous, framing-robust negative result we aim to temper deployment claims. No human subjects or private data are involved; product references are anonymized.

Software and Data

The persona-simulation harness, prompts, anonymized case corpus, and per-persona logs for all framings are released as part of the open-source SoloSquad project (SoloSquad, 2026) at https://github.com/Adelie-Squad/solosquad (logs under reports/logs/). Author identity is withheld for review.

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A. Anonymized Case Corpus

A device-syncing tool via demo-video waitlist (real: success); B online shoe store via concierge fake store (success); C reformulated flagship drink that won blind taste tests, replacing the original for loyal customers (failure); D voice dictation assessed by Wizard-of-Oz for sustained use (low demand); E social-post scheduler via landing-page pricing (success). Ground truth is revealed only for scoring.

B. Framings, Prompts, and Targets

Stated elicits would_pay; revealed elicits retained_pay after a click→try→retain funnel with incumbent attachment; graded elicits pay_prob$\in [0, 1]$. Targets were fixed a priori by keyword (e.g., E: marketing/creator/self-employed terms) matched against occupation and persona-narrative fields. Matched target counts (8000-pool): A 3062, B 711, C 2034, D 2834, E 1023. Full prompts and keyword lists are in the repository.

C. Per-Framing Logs

Per-persona decisions and free-text reasons are in results_{stated,revealed,graded}_{general,target}.json and, for real products, results_products_graded_*.json.